Tzu-Hsiang (Sean) LIN

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EDUCATION

University of Michigan **Ann Arbor, MI, USA**
Master's of Science in Mechanical Engineering (Robotics) *2025-2027*
Coursework: *Mathematics for Robotics (ROB501), Robotics System Lab (ROB550), Mobile Robot: Algorithm (ROB530)*
Research Lab: DASC Lab; Zoetic Robotics Lab; SAM Lab

University of Nottingham **Nottinghamshire, UK**
Bachelor's of Engineering (Hons) in Mechanical Engineering *2021-2025*
GPA: 3.90/ 4.00 (Top1%)
Honors: Nottingham Global **Scholarship**, Dean's **Scholarship,** Ningbo MoE **Scholarship**, Ningbo Government **Scholarship**

SKILLS

Mechanical: CAD (SolidWorks, CATIA, NX) FEA, 3D-printing, CFD, GD&T, DFM, 3D modelling, engineering drawing, machining
Programming & Simulation: Python, C, C++, Git, Linux, ANSYS, Abaqus, DS&A, MATLAB, Simulink, OpenCV, RViz2, Gazebo
Robotics & Control: SLAM, ROS2, ROS, Sim2Real, NVIDIA Isaac Sim, YOLO, PyTorch, GNN, Kalman filters, LQR, MPC, A*
Mechatronics: Arduino, Raspberry Pi, STM32, ESP32, IMU, LiDAR, RGB-D, Sensor Integration, pneumatic actuators, electronics

WORK EXPERIENCE

Apple Inc.
Product Design Engineering Intern *May. 2026- Sep. 2026*
- Supported Apple Watch NPI builds through FA, TA, hardware validation, and DOE during production ramp.
- Investigated design and manufacturing issues to improve product reliability through cross-functional engineering collaboration including operation team and EE team.
- Developed an LLM-assisted engineering tool to tolerance analysis and reduce engineering workflow time by *87%*.
- Designed a control algorithm for an automated water-seal testing fixture, reducing test cycle time by *91%*.
- Developed a computer-vision dimensional inspection pipeline for improving the FA process

Bellwether Industries
Mechanical Engineer Intern *Aug. 2024- Oct. 2024*
- Designed and integrated mechanical, propulsion, and electronic systems for a *1/8-scale eVTOL prototype*.
- Built a multi-orientation flight-dynamics test platform for vehicle testing and propulsion evaluation.
- Developed MATLAB/Simulink/PX4 simulation workflows for eVTOL system analysis and validation.
- Performed *CFD and FEA analysis* to evaluate aerodynamic performance and structural loading.

MECHANICAL PROJECT EXPERIENCES

Variable Pitch Propeller / Engineering Design / University of Nottingham **Nottingham, UK**
Mechanical Design/ Simulation Engineer *Jan. 2023 - Jun. 2023*
- Applied Design for Manufacturing and Design for Assembly principles to reduce cost for mass production feasibility.
- Designed a Hydraulic system and a Scotch-York mechanism to achieve linear-to-rotation motion with SolidWorks.
- Optimized design of critical components with the result of a safety factor greater than 2.5.

Formula Student Racing / Notts Racing / University of Nottingham **Nottingham, UK**
Aerodynamic Engineer/ Mechanical Design Engineer *Oct. 2022 - Jan. 2025*
- Designed aerodynamic and structural components using CATIA to improve vehicle performance and packaging.
- Conducted CFD analysis of aerodynamic components and cooling systems for vehicle optimization.
- Designed pedal-box systems and evaluated hydraulic braking performance using Simulink/Simscape.

ROBOTICS PROJECT EXPERIENCES

Mobile Robot Perception, Navigation and Control/ Robotics System Lab / University of Michigan **Ann Arbor, MI**
Robotics Engineer *Oct. 2025 - Present*

- Developed a C++/ROS2 autonomy stack integrating LiDAR, IMU, encoders, localization, planning, and PID control.
- Implemented occupancy-grid mapping, Monte Carlo Localization, and A* for autonomous navigation.
- Achieved 10 Hz real-time particle-filter localization with thousands of particles on embedded hardware.

5-DoF Robotics Arm Autonomy / Robotics System Lab / University of Michigan **Ann Arbor, MI**
Robotics Engineer *Sep. 2025 - Nov. 2025*

- Integrated computer vision sensing and kinematic reasoning for a 5-DOF robotic arm with an Intel RealSense camera.
- Applied intrinsic and extrinsic calibration for precise vision alignment and enabled semi-autonomous pick-and-place
- Coding Python file block detection via OpenCV and Pox-based kinematics, achieving 1.9% average positional error.

RESEARCH EXPERIENCE

University of Michigan / Zoetic Robotics Lab **Ann Arbor, MI**
Robotics Researcher *Sep. 2025 – Present*

- Developed experimental instrumentation and data-acquisition workflows for pneumatically actuated *bio-inspired aerial robots*, involving in designing, simulation and mathematical modelling.
- Built image-processing and tracking workflows to quantify small actuator displacements from high-speed video and characterize dynamic response across operating conditions.
- Developed physics-based and semi-empirical models of nonlinear actuator behaviour and compared simulation predictions against experimental measurements.

University of Michigan / Distributed Autonomous Systems and Control Lab **Ann Arbor, MI**
Robotics Researcher *Jan. 2026 – Present*

- Developed and deployed safety-critical control algorithms for multi-agent aerial robots, integrating CBF-MPC and gatekeeper-based safety mechanisms for collision-constrained exploration.
- Implemented and validated control and autonomy modules on *Crazyflie hardware*, supporting both single-robot and multi-agent experiments in GPS-denied environments.
- Integrated planning, control, and motion-capture systems through custom ROS interfaces and coordinate-frame transformations**,** enabling closed-loop execution from simulation to physical hardware.
- Investigated real-world failure modes including controller instability, state-estimation error, communication latency, and model mismatch during close-proximity multi-robot flight.
- Built an Optuna-based hyperparameter tuning and benchmarking pipeline for BR-MPPI, optimizing controller parameters and evaluating safety, trajectory-tracking performance, and computational efficiency.

University of Nottingham / Advanced Robotics Laboratory **Nottingham, UK**
Mechanical (Robotics) Researcher *Aug. 2024 – Jun. 2025*

- Developed a bio-inspired manta-ray soft robot using pneumatic actuators and Piecewise Constant Curvature kinematics.
- Implemented MATLAB-based locomotion modelling and actuator analysis for robotic motion control.
- Fabricated and experimentally validated soft pneumatic actuators under varying pressure conditions.

University of Nottingham / Robotics Laboratory **Nottingham, UK**
Mechanical (Robotics) Researcher *Aug. 2024 – Jun. 2025*

- Conducting in-depth analyses in bladeless drone fluid mechanics, comparing with traditional systems.
- Apply advanced simulation techniques using ANSYS to model fluid dynamics on the single-motor bladeless drone and Dyson fan performance.

PUBLICATION

- S. Kim*, Y.-H. Hsiao*, A. Yin*, **T.-H. Lin**, J. Brown, G. Matt, Y. Chen, C. Aubin, **Controlled hovering flight in a sub-gram micro aerial vehicle powered by pneumatic micro-muscles** *(in prep)* (2025)
- T. Kim*, R.H. Kumar*, A.D. Menon, **T.-H. Lin**, D. Panagou**, SEAMLiS: Visibility-Aware Safety for Perception-Limited Multi-Robot Exploration**, *arXiv preprint, 2026*. [Project Page] [arXiv]